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Kalman smoothing improves the estimation of joint kinematics and kinetics in marker-based human gait analysis

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ABSTRACT

We developed a Kalman smoothing algorithm to improve estimates of joint kinematics from measured marker trajectories during motion analysis. Kalman smoothing estimates are based on complete marker trajectories. This is an improvement over other techniques, such as the global optimisation method (GOM), Kalman filtering, and local marker estimation (LME), where the estimate at each time instant is only based on part of the marker trajectories. We applied GOM, Kalman filtering, LME, and Kalman smoothing to marker trajectories from both simulated and experimental gait motion, to estimate the joint kinematics of a ten segment biomechanical model, with 21 degrees of freedom. Three simulated marker trajectories were studied: without errors, with instrumental errors, and with soft tissue artefacts (STA). Two modelling errors were studied: increased thigh length and hip centre dislocation. We calculated estimation errors from the known joint kinematics in the simulation study. Compared with other techniques, Kalman smoothing reduced the estimation errors for the joint positions, by more than 50% for the simulated marker trajectories without errors and with instrumental errors. Compared with GOM, Kalman smoothing reduced the estimation errors for the joint moments by more than 35%. Compared with Kalman filtering and LME, Kalman smoothing reduced the estimation errors for the joint accelerations by at least 50%. Our simulation results show that the use of Kalman smoothing substantially improves the estimates of joint kinematics and kinetics compared with previously proposed techniques (GOM, Kalman filtering, and LME) for both simulated, with and without modelling errors, and experimentally measured gait motion.

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1. Introduction

Inverse kinematics, the estimation of joint kinematics based on measured trajectories of skin-mounted markers, is complicated by instrumental errors and soft tissue artefacts (STA) (Cappozzo et al., 1996; Chiari et al., 2005; Leardini et al., 2005).

Different techniques to reduce the effect of these errors on the estimated joint kinematics have been proposed (Chiari et al., 2005; Leardini et al., 2005). Spoor and Veldpaus (1980) estimated the positions and orientations of each body segment separately using a segmental optimisation method (SOM). SOM minimises the marker displacement in the segmental reference frame between any two time instants. Lu and O'Connor (1999) used a multi-link model relating the marker positions to the generalized co-ordinates that describe the motion of the body segments along the degrees of freedom (DOFs). At each time instant, their global optimisation method (GOM) estimates all generalized co-ordinates at once from a weighted nonlinear least-squares fit between the measured marker positions and those predicted by the model. GOM outperformed SOM in simulation for a serial three-link model (pelvis, thigh, and shank) joined by two spherical joints (hip and knee), suggesting that imposed joint constraints reduce the effect of errors. Cerveri et al. (2003a, b) used a Kalman filter to estimate joint kinematics. Kalman filtering (KF) is based on a measurement model obtained from the biomechanical model and a process model, which includes prior knowledge about the smoothness of the motion. In addition, the generalized co-ordinates, velocities, and accelerations are estimated simultaneously. Cerveri et al. (2005) proposed local marker estimation (LME), an extension of KF to estimate marker displacements in the segmental reference frames to account for STA. In their simulation study (Cerveri et al., 2005) in which systematic, sinusoidal perturbations added to the three thigh markers modelled STA, LME estimates were at least 50% more accurate than SOM estimates.

KF has two potential advantages over GOM. Firstly, including knowledge about motion smoothness may improve the accuracy of estimated joint kinematics. Secondly, estimating accelerations

Abbreviations: DOF, degree of freedom; GOM, global optimisation method; KF, Kalman filtering; KS, Kalman smoothing; LME, local marker estimation; RMS, root mean square; SOM, segmental optimisation method; STA, soft tissue artefacts

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eliminates the need to differentiate generalized co-ordinates numerically, which can introduce large errors. As these accelerations, in addition to the measured ground reaction forces, are the input for inverse dynamics to calculate joint moments, more accurate joint acceleration estimates will improve the accuracy of joint kinetics. Since Cerveri et al. (2005) did not compare LME with GOM, these advantages have not yet been confirmed.

A drawback of KF is the asymmetrical use of data. At each time instant, estimates are based on the measured marker trajectories up to the considered time instant only. We therefore propose Kalman smoothing (KS) (Rauch et al., 1965), a combination of two filters, to calculate the estimates at each time instant based on the complete marker trajectories. The proposed KS is an extension of the Kalman filter without LME. The purpose of this study was to compare the accuracies of the generalized co-ordinates and accelerations using GOM, KF, LME, and KS using both simulated marker trajectories, with and without modelling errors, and experimentally measured marker trajectories during gait.

2. Kalman Smoothing algorithm

The Kalman smoother (Bar-Shalom and Li, 1993; Rauch et al., 1965) combines prior knowledge, described by a process and measurement model, with the measured marker trajectories to produce an estimate of the joint kinematics while minimising the estimation error statistically. The generalized co-ordinates q and their derivatives up to the Kth order, which describe the joint kinematics, are collected in a vector x:

$$x = [q_1 q_1^{(1)} \dots q_1^{(K)} q_2 q_2^{(1)} \dots q_2^{(K)} \dots q_J q_J^{(1)} \dots q_J^{(K)}]^{\mathrm{T}},$$

with $j = 1 \dots J$ indicating the DOF and $q_j^{(k)}$ the kth time derivative of q_j.

The process model describes the expected time evolution of the joint kinematics x and is composed of J submodels describing the motion of each DOF. While the submodels are based on the assumption that the Kth derivative of the generalized co-ordinate $q_j^{(K)}$ is constant, a noise term n_j takes into account the errors introduced by this assumption:

$$\begin{bmatrix} q_j(t_i+\Delta t) \\ q_j^{(1)}(t_i+\Delta t) \\ \vdots \\ q_j^{(K)}(t_i+\Delta t) \end{bmatrix} = \begin{bmatrix} 1 & \Delta t & \frac{\Delta t^2}{2} & \frac{\Delta t^K}{K!} \\ 0 & 1 & \Delta t & \frac{\Delta t^{K-1}}{(K-1)!} \\ 0 & & & 1 \end{bmatrix} \begin{bmatrix} q_j(t_i) \\ q_j^{(1)}(t_i) \\ \\ q_j^{(K)}(t_i) \end{bmatrix} + n_j(t_i),$$

with t is the time, $I = 1 \dots I$ indicates the time instant, and Δt is the sample time. The (K+1)th derivative of the generalized co-ordinate is modelled as zero mean Gaussian noise with covariance $\sigma^2_{K+1,j}$. Therefore, the process noise is given by

$$n_j(t) = N(0, Q_j) \quad \text{with} \quad Q_j = \sigma^2_{K+1,j} G^{\mathrm{T}} G \text{ and}$$
$$G = \left[\frac{\Delta t^{K+1}}{(K+1)!} \quad \frac{\Delta t^K}{K!} \quad \dots \quad \Delta t \right].$$

The measurement model relates the joint kinematics $x(t)$ to the measured marker positions, collected in $z(t)$. This model is composed of a noiseless measurement model $h(x(t))$ and measurement noise $v(t)$:

$$z(t) = h(x(t)) + v(t)$$

The noiseless measurement model $h(x(t))$ is based on a biomechanical model consisting of ten body segments including 21 DOFs (Fig. 1, Delp et al., 1990). The measurement noise, $v(t)$, is drawn from a zero mean Gaussian distribution and expresses the uncertainty for the marker position measurements.

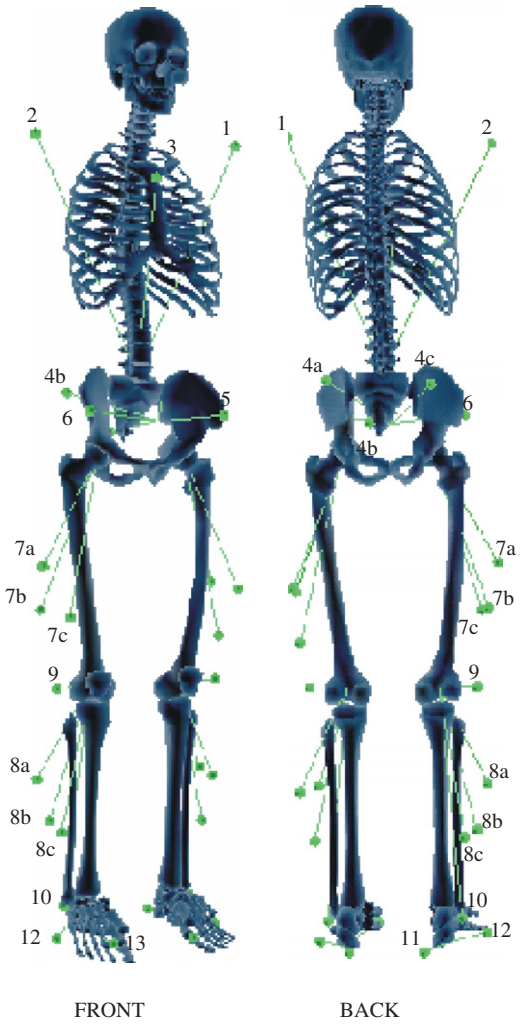


Fig. 1. Biomechanical model and marker placement protocol. The biomechanical model consists of ten body segments: a head–arms–trunk segment, the pelvis, left and right thigh, shank, hindfoot and forefoot (Delp et al., 1990). This model includes 21 DOFs. Spherical joints connect the head–arms–trunk segment to the pelvis and the pelvis to the thighs. The ankle and subtalar joints are modelled as simple hinges, whereas the knee joints are modelled as sliding hinges (Yamaguchi and Zajac, 1989). The remaining six DOFs correspond to the position and orientation of the pelvis. The generic biomechanical model was scaled to the subject's dimensions. A modified Cleveland marker placement protocol (Sutherland, 2002) was used for the data collection. The marker set consisted of 30 markers, including five clusters of three markers. Three anatomical markers defined the trunk: a marker on the lateral aspects of the left (1) and right (2) shoulder and a marker on the sternum (3). The pelvis segment is defined by a cluster of three technical markers on the sacrum (4a–c) and two anatomical markers on the left (5) and right (6) anterior superior iliac spine (ASIS). The thigh segment is defined by a cluster of three technical markers (7a–c). The shank segment is defined by a cluster of three technical markers (8a–c), an anatomical marker on the lateral epicondyle (9), and an anatomical marker on the lateral malleolus (10). The foot segment is defined by three anatomical markers on the heel (11), the lateral foot (12) and the first metatarsal head (13). During a static calibration trial, additional anatomical markers were added to the medial femoral condyles and the medial malleoli to define the knee and ankle joint axis.

KS has two consecutive steps. First, a Kalman filter (Kalman, 1960) estimates the joint kinematics at t_i using only the measured marker trajectories up to t_i. Second, a backward recursion using the measured marker trajectories from the last instant down to t_i, follows the Kalman filter. The resulting Kalman smoother estimates the generalized co-ordinates and their derivatives based on all the information available: the complete marker trajectories, the process model, and the measurement model. An extended

smoother (Bar-Shalom and Li, 1993) is used to cope with the nonlinearity of the measurement model.

3. Validation

We compared the kinematics estimated with GOM, KF, LME, and KS, first using simulated marker trajectories and second using experimentally measured marker trajectories both sampled at 200 Hz. The marker set comprised 30 markers including five clusters of three markers (Fig. 1).

3.1. Validation in simulation

We used the biomechanical model to calculate three sets of marker trajectories from the kinematics of a gait motion described by generalized co-ordinates q_{gt} (ground truth): (1) ideal marker trajectories without errors, consistent with q_{gt} and the biomechanical model; (2) marker trajectories with instrumental errors, obtained by adding zero mean Gaussian noise with standard deviation of 1 mm to the ideal marker trajectories; (3) marker trajectories corrupted with STA calculated as explained below.

STA are related to the angles of adjacent joints (Cappozzo et al., 1996). We calculated such a relation from marker trajectories measured experimentally during gait. Inverse kinematics (SIMM, Motion Analysis Corporation) estimated the generalized co-ordinates $q_{exp}(t)$ corresponding to these trajectories. Using the biomechanical model, the measured marker positions were projected in the segmental reference frame. We considered the time dependent distances between the projected and modelled marker positions to be an approximation of the STA. For each marker, this distance, $d(t)$ was calculated along the x-, y-, and z-direction of the segmental reference frame and a linear relation between this distance and the adjacent joint angles $q_{exp,l}$ was calculated using least-squares:

$$\min_a \left[\sum_{i=1}^{I} \left(d(t_i) - \sum_{l=1}^{L} a_l q_{exp,l}(t_i) \right)^2 \right],$$

with $l = 1 \ldots L$ indicating the adjacent joints. The resulting coefficients a_l were used to calculate the modelled STA as

$$\sum_{l=1}^{L} a_l q_{gt,l}.$$

The modelled STA (Fig. 2) had amplitudes between 5 and 25 mm. It should be noted that, in contrast to the observations of Cappozzo et al. (1996), the x- and z-components of the STA trajectories in the shank and foot are correlated in our data set.

We applied GOM, KF, LME, and KS to the simulated marker trajectories. All methods were based on the biomechanical model of Delp et al. (1990). The algorithm for inverse kinematics implemented in SIMM (Delp et al., 1990) calculated the GOM estimates according to Lu and O'Connor (1999). SIMM uses the Levenberg–Marquardt algorithm (Moré, 1977) to solve the underlying nonlinear least-squares problem with the estimated generalized co-ordinates at the previous time instant as initial values. KF used a third-order process model. LME was applied to all 30 markers. As recommended by Cerveri et al. (2005), we used a second-order process model and the covariance associated with the local marker co-ordinates was two orders of magnitude smaller than those associated with the generalized co-ordinates. To investigate the influence of the order of the process model, we performed KS for $K = 2$, $K = 3$, and $K = 4$.

The effect of modelling errors was investigated by applying GOM, KF, and KS to the ideal marker trajectories using a biomechanical model that differed from the model used to simulate marker trajectories. The studied modelling errors were a thigh length increased by 10% and a dislocation of the joint centre of the hip of 16.8 mm.

For all estimates, we calculated the marker error, i.e. the root mean square (RMS) distance between the simulated marker trajectories and those predicted by the estimates and the biomechanical model averaged over all markers, $\bar{d}_{RMS}$.

Thereafter we calculated the percentage error for the generalized co-ordinates averaged over the 21 DOFs, ε_{pos}:

$$\varepsilon_{pos} = \frac{1}{J} \sum_{j=1}^{J} 100 \frac{\sqrt{(1/I) \Sigma_{i=1}^{I} \left(q_j(t_i) - q_{gt,j}(t_i) \right)^2}}{\max(q_{gt,j}) - \min(q_{gt,j})},$$

where q_j is the considered position estimate of co-ordinate j, $\min(q_{gt,j})$ and $\max(q_{gt,j})$ are, respectively, the extreme values of co-ordinate j during gait.

Likewise, we calculated the percentage errors for the generalized accelerations, ε_{acc}, for KF, LME, and KS.

Since GOM does not yield acceleration estimates, calculating generalized accelerations and joint moments requires double differentiation of the position estimates. To study the effect of numerical differentiation on the quality of the joint moment calculation, we calculated joint moments using SIMM dynamics pipeline and SD Fast (Motion Analysis Corporation, PTC Inc.) for three different joint kinematics: the ground truth, the estimates obtained with GOM, and those obtained with third-order KS. Ground reaction forces, consistent with q_{gt}, were input to the joint moment calculation. The calculation of the percentage errors for the joint moments, ε_T, is analogous to the calculation of ε_{pos}. We compared percentage errors for the extreme moments, $\varepsilon_{T\ min,j}$ and $\varepsilon_{T\ max,j}$, for GOM and KS:

$$\varepsilon_{T\,min,j} = 100 \frac{|\min(T_j) - \min(T_{gt,j})|}{\max(T_{gt,j}) - \min(T_{gt,j})} \quad \text{and}$$

$$\varepsilon_{T\,max,j} = 100 \frac{|\max(T_j) - \max(T_{gt,j})|}{\max(T_{gt,j}) - \min(T_{gt,j})},$$

where T_j and $T_{gt,j}$ are the moments of joint j, calculated based on the estimates and the ground truth, respectively.

3.2. Experimental validation

We applied GOM, KF, and KS to experimentally measured marker trajectories. Marker trajectories were measured during gait using a seven camera motion capture system (Qualisys Motion Capture systems). We could not compare estimation errors in this part of the study because no ground truth for the joint kinematics was known. We therefore assumed that improved estimates of the generalized co-ordinates would better predict the trajectories of validation markers. The trajectories of validation markers are measured but are not used to estimate the generalized co-ordinates. To validate our method based on this assumption, estimates of the generalized co-ordinates were calculated from a subset of the measured markers including only one marker on the thigh, while the other two markers on the thigh were used as validation markers. The positions of the validation markers were calculated from the estimates of the generalized co-ordinates using the biomechanical model. Then, the root RMS distance between the measured and calculated positions of the validation markers, d, was used as a measure of the estimation quality.

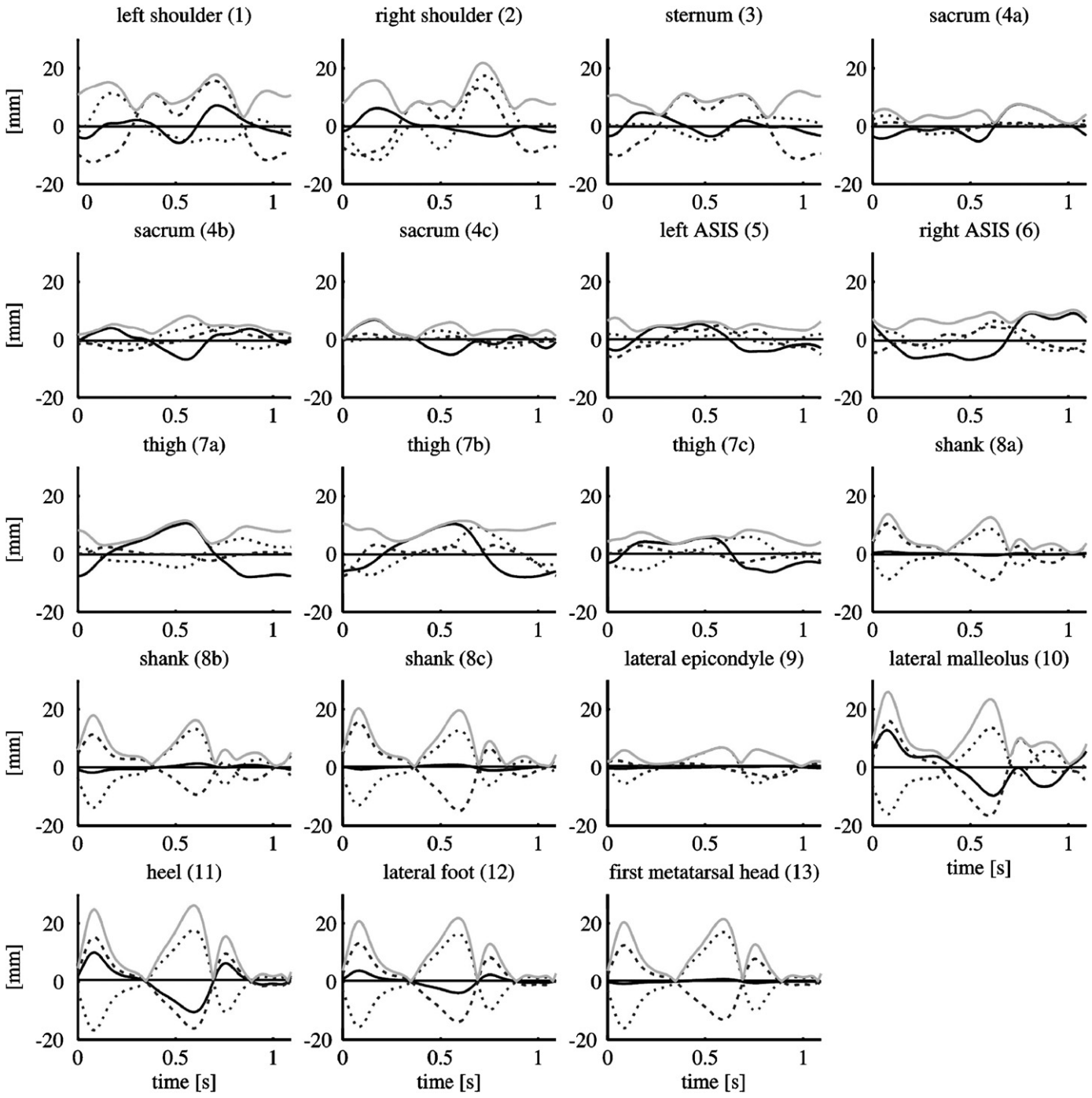


Fig. 2. Time trajectories of the simulated STA for the markers on the trunk, pelvis and right leg: *x*-component (dotted black), *y*-component (solid black), *z*-component (dashed black) in the segmental reference frame and magnitude (gray). Marker numbering is consistent with Fig. 1.

4. Results

4.1. Simulation results

The smallest marker errors, $\bar{d}_{\mathrm{RMS}}$, were obtained with LME, the largest marker errors were obtained with GOM (Table 1).

KS resulted in a smaller estimation error for the generalized co-ordinates, $\varepsilon_{\mathrm{pos}}$, than GOM, KF, and LME (Table 2). Compared with GOM and KF respectively, KS ($K = 3$) reduced the estimation error by 97.8% and 59.3% (without errors), 88.1% and 53.0% (instrumental errors), 4.5% and 0.2% (STA), 6.5% and 0.9% (increased thigh length), and 10.8% and 0.1% (hip centre dislocation). Percentage reductions are calculated from the estimation errors (supplementary material). The errors for the LME position estimates were two to three orders of magnitude larger than the errors with the other techniques.

The effect of the order K of the process model of KS on the estimation errors depended on the errors of the input marker trajectories. A second-order process model outperformed third- and fourth-order process models for marker trajectories without errors. In the presence of instrumental errors, third- and even fourth-order process models produced better estimates than a second-order process model. In the presence of STA, the estimation errors were similar for all three orders (Table 2).

Table 1
Marker error, which is the root mean square (RMS) distance between the marker trajectories calculated based on the estimates of the generalized co-ordinates and the simulated marker trajectories averaged over all markers, $\bar{d}_{RMS}$.

Marker error, $\bar{d}_{RMS}$ (mm)	GOM	LME	Kalman filter ($K=3$)	Kalman smoother ($K=2$)	Kalman smoother ($K=3$)	Kalman smoother ($K=4$)
Without errors	10.27	0.01	0.04	0.01	0.03	0.08
Instrumental errors	10.52	0.10	1.45	1.55	1.59	1.58
Soft tissue artefacts	9.57	0.01	3.46	3.46	3.46	3.46
Modelling errors						
Increased thigh length	23.40		15.19		15.19	
Hip centre dislocation	12.82		3.60		3.60	

Table 2
Percentage errors for the estimates of the generalized co-ordinates, ε_{pos}, obtained with different methods from simulated marker trajectories without errors, with instrumental errors, and with soft tissue artefacts (STA) using a biomechanical model without modelling errors, and from simulated marker trajectories without errors using a biomechanical model with an increased thigh length and a hip centre dislocation.

Percentage error for the estimates of the generalized co-ordinates, ε_{pos}	GOM	LME	Kalman filter ($K=3$)	Kalman smoother ($K=2$)	Kalman smoother ($K=3$)	Kalman smoother ($K=4$)
Without errors	4.045	204	0.209	**0.030**	0.085	0.215
Instrumental errors	4.785	585	1.203	0.621	**0.565**	0.568
Soft tissue artefacts	7.120	541	6.814	**6.798**	**6.798**	6.799
Modelling errors						
Increased thigh length	25.69		24.23		**24.02**	
Hip centre dislocation	13.01		11.61		**11.60**	

The best estimate is indicated in bold.

Table 3
Percentage errors for the estimates of the generalized accelerations, ε_{acc}, obtained with different methods from simulated marker trajectories without errors, with instrumental errors, and with soft tissue artefacts (STA) using a biomechanical model without modelling errors, and from simulated marker trajectories without errors using a biomechanical model with an increased thigh length and a hip centre dislocation.

Percentage errors for the estimates of the generalized accelerations, ε_{acc}	LME	Kalman filter ($K=3$)	Kalman smoother ($K=2$)	Kalman smoother ($K=3$)	Kalman smoother ($K=4$)
Without errors	25.93	16.10	**1.77**	2.68	4.01
Instrumental errors	100.12	37.50	20.00	7.29	**6.46**
Soft tissue artefacts	53.21	20.53	**9.42**	9.57	9.95
Modelling errors					
Increased thigh length		42.75		**18.84**	
Hip centre dislocation		17.85		**4.85**	

The best estimate is indicated in bold.

KS resulted in a smaller estimation error for the generalized accelerations, ε_{acc}, than KF and LME (Table 3). Compared with KF, KS ($K=3$) reduced the estimation error by 83.3% (without errors), 80.6% (instrumental errors), 53.4% (STA), 55.9% (increased thigh length), and 72.8% (hip centre dislocation). The KF estimates showed a time lag with respect to the ground truth (Fig. 3). This time lag was larger for higher derivatives of the generalized co-ordinates and disappeared when smoothing was applied. Compared with LME, KS reduced estimation errors by one to two orders of magnitude.

Compared with GOM, KS reduced the estimation error for the joint moments, ε_T, by 87.7% (without errors), 85.1% (instrumental errors), 45.6% (STA), 36.2% (increased thigh length) and 57.4% (hip centre dislocation) (Table 4, Fig. 4). KS estimated more accurate extreme joint moments than GOM (Fig. 5).

4.2. Experimental results

Marker positions calculated using KS showed higher correspondence with measured marker positions, than those calculated with GOM (Table 5, Fig. 6). Compared with GOM, KS reduced the RMS distance between measured and calculated marker positions by 16.2% for the thigh marker used for estimation and by 15.5% and 24.3% for the two validation markers. RMS distances resulting from KF and KS were similar (Table 5).

5. Discussion

Several methods to reduce the sensitivity of inverse kinematics to instrumental errors and STA, have been proposed in the

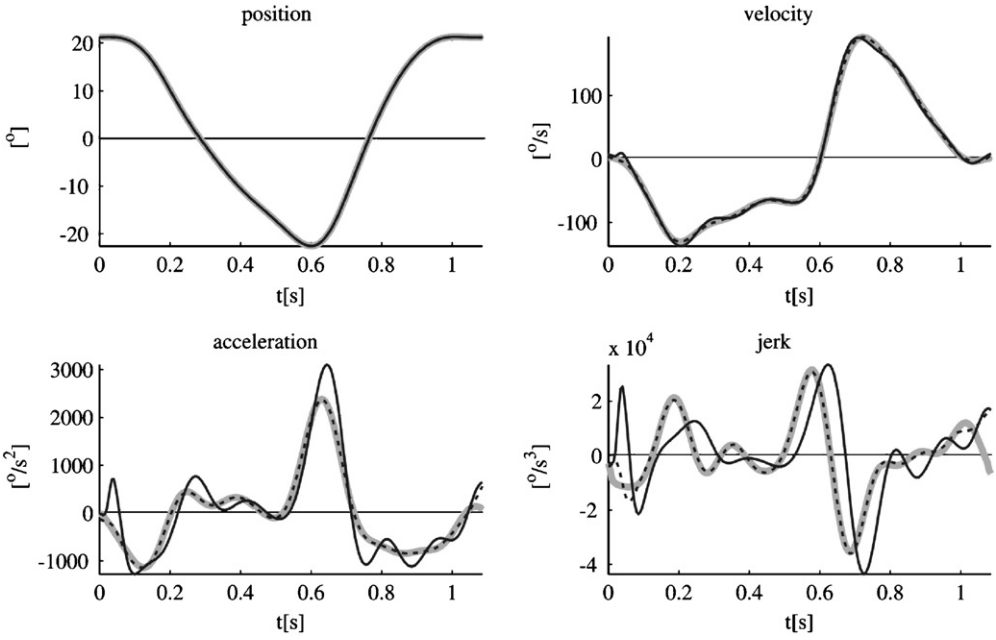


Fig. 3. Estimates of angular position, velocity, acceleration and jerk of the right hip flexion as a function of time obtained from the set of marker trajectories without errors with KF (solid black) and smoothing (dashed black). The ground truth (gray) serves as a reference. The filter estimate suffers from a time lag.

Table 4
Percentage errors for the joint moments, ε_T, calculated based on the estimates of the generalized co-ordinates obtained with the global optimisation method (GOM) and Kalman smoothing from simulated marker trajectories without errors, with instrumental errors, and with STA using a biomechanical model without modelling errors, and from simulated marker trajectories without errors using a biomechanical model with an increased thigh length and a hip centre dislocation.

Percentage errors for the joint moments ε_T	GOM	Kalman smoother ($K = 3$)
Without errors	12.17	1.50
Instrumental errors	12.24	1.82
Soft tissue artefacts	12.23	6.65
Modelling errors		
Increased thigh length	14.19	9.06
Hip centre dislocation	13.43	5.72

literature: SOM, GOM, KF, and LME. A drawback of these methods is that they only use part of the marker trajectories to estimate the joint kinematics at a considered time instant. We therefore developed a KS algorithm to calculate the estimate at each time instant based on the complete marker trajectories. To assess whether KS improves estimates of joint kinematics and kinetics, we compared this technique with GOM, KF, and LME. We did not include SOM in our study, since GOM and LME have already been shown to outperform SOM (Cerveri et al., 2005; Lu and O'Connor, 1999).

KS resulted in better estimates of the generalized co-ordinates calculated from simulated marker trajectories than GOM, KF, and LME. Furthermore, KF outperformed GOM in terms of estimation errors (Tables 2 and 3). The smaller estimation errors using KF and smoothing compared to GOM confirm our assumption that inverse kinematics benefits from integrating prior knowledge on motion smoothness.

Although GOM narrowly focuses on minimising marker error at a given time instant, KF and KS outperform GOM in the reduction of marker error (Table 1). In the absence of measurement and modelling errors (validation in simulation with the marker trajectories without errors), a perfect fit between the measurements and the biomechanical model exists. Nevertheless, the obtained marker error for the GOM estimate is 10.27 mm, showing that GOM does not return the global optimum. It is a well-known problem that solvers for nonlinear (nonconvex) optimisation problems find a locally optimal solution depending on the initial values of the optimisation variables. Since in this study the initial values at the first time instant were equal for all methods, this shows that KF and KS are more suitable to solve this problem.

Cerveri et al. (2005) improved the estimation of joint kinematics by applying LME to three thigh markers. However, our results (Tables 2 and 3) show that LME is not suitable to cope with STA of the full marker set during gait. Estimating all 30 local marker positions, velocities, and accelerations leads to a huge number of variables in the underlying Kalman filter. Compared to the simulation study of Cerveri et al. (2005), the joint range of motion during gait and therefore the information contained in the marker positions is limited. Since this information is insufficient to correctly estimate all local marker positions, the estimated joint kinematics diverge from the ground truth, although LME outperforms all other methods in terms of marker error, $\bar{d}_{RMS}$ (Table 1).

Simulation results (Tables 2 and 3) showed that in the presence of instrumental errors, a third- or fourth-order process model outperformed a second-order process model for the Kalman smoother.

We addressed the effect of KF and KS on the accuracy of the estimated accelerations. While the Kalman smoother estimates of the generalized co-ordinates where twice as accurate (without errors, instrumental errors) or as accurate (STA, modelling errors) as the corresponding Kalman filter estimates (Table 2), the Kalman smoother estimates of the generalized accelerations were respectively about five times (without errors, instrumental errors) and about two times (STA, modelling errors) better than the corresponding Kalman filter estimates (Table 3). The inferior

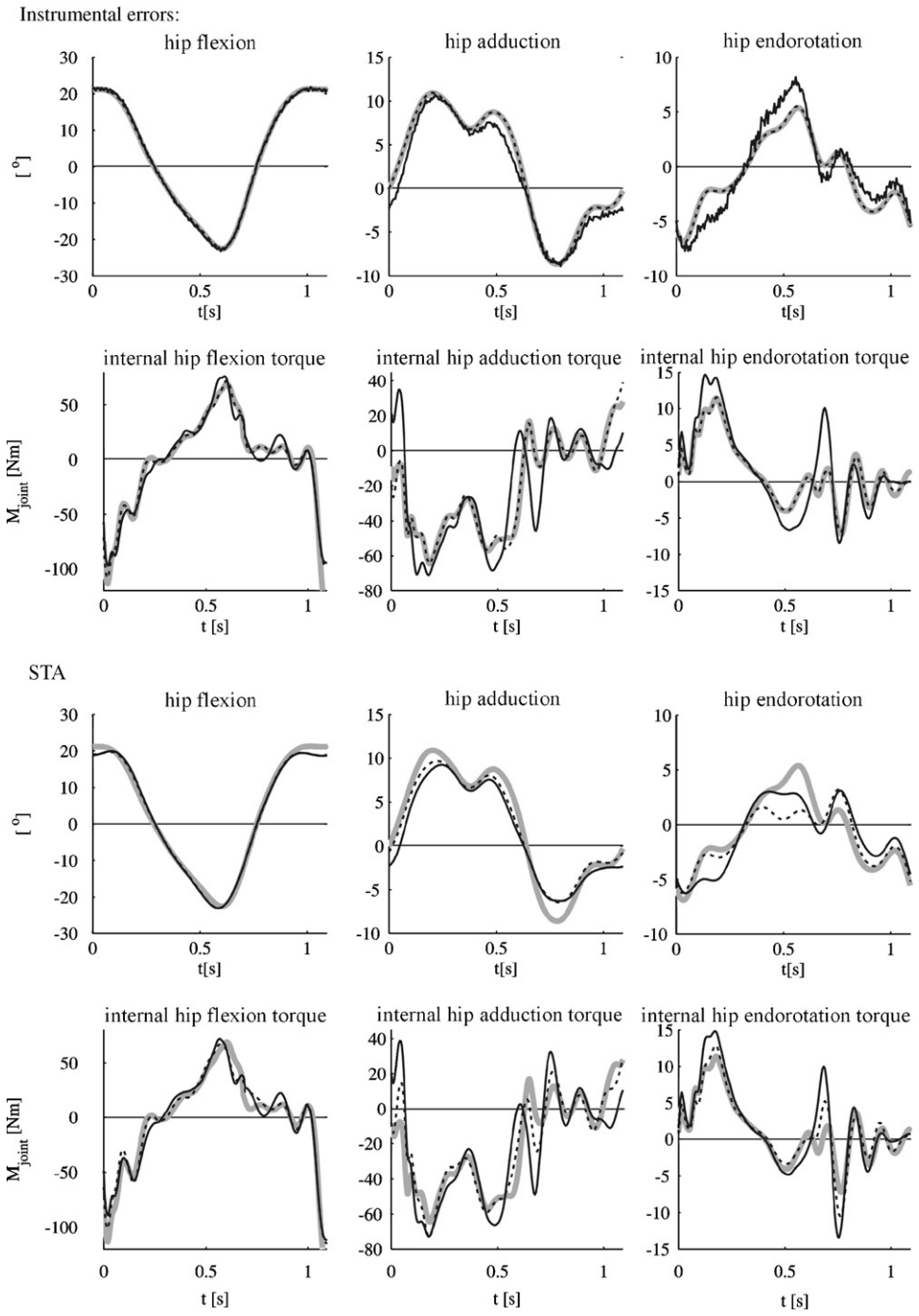


Fig. 4. Estimates of angular positions and internal joint moments of the right hip as a function of time obtained from the set of marker trajectories with instrumental errors above, and with STA, below. Estimates obtained with the GOM are represented by a solid black line, estimates obtained with third-order KS are represented by a dashed black line. The ground truth (gray) serves as a reference.

acceleration estimates of KF were due to the filter lag (Fig. 3) caused by the asymmetrical use of data in time. The backward recursion added by the Kalman smoother not only improves the velocity and acceleration estimates but, through the process model, also consistently improves the position estimates.

Using KS instead of GOM greatly improved the estimation of joint moments and consequently the assessment of maximum joint loading from peak moments even when STA or modelling errors were present and the reduction of the estimation errors for the positions was rather limited. The inferior joint moment estimation of GOM results from the numerical differentiation of the nonsmooth positions. GOM associated with a double differentiation technique does not yield results that are as good as those obtained with KF or KS since such a technique only finds accelerations corresponding to the estimated positions but does not improve the position estimates. The advantage of KF and KS relies on the use of a process model that links one time instant to the next reflecting the knowledge that the state of the system is not independent of the states at neighbouring time instances.

GOM, KF and KS were applied to marker trajectories measured experimentally during gait. The estimates obtained with KF and smoothing better explained the marker trajectories and better

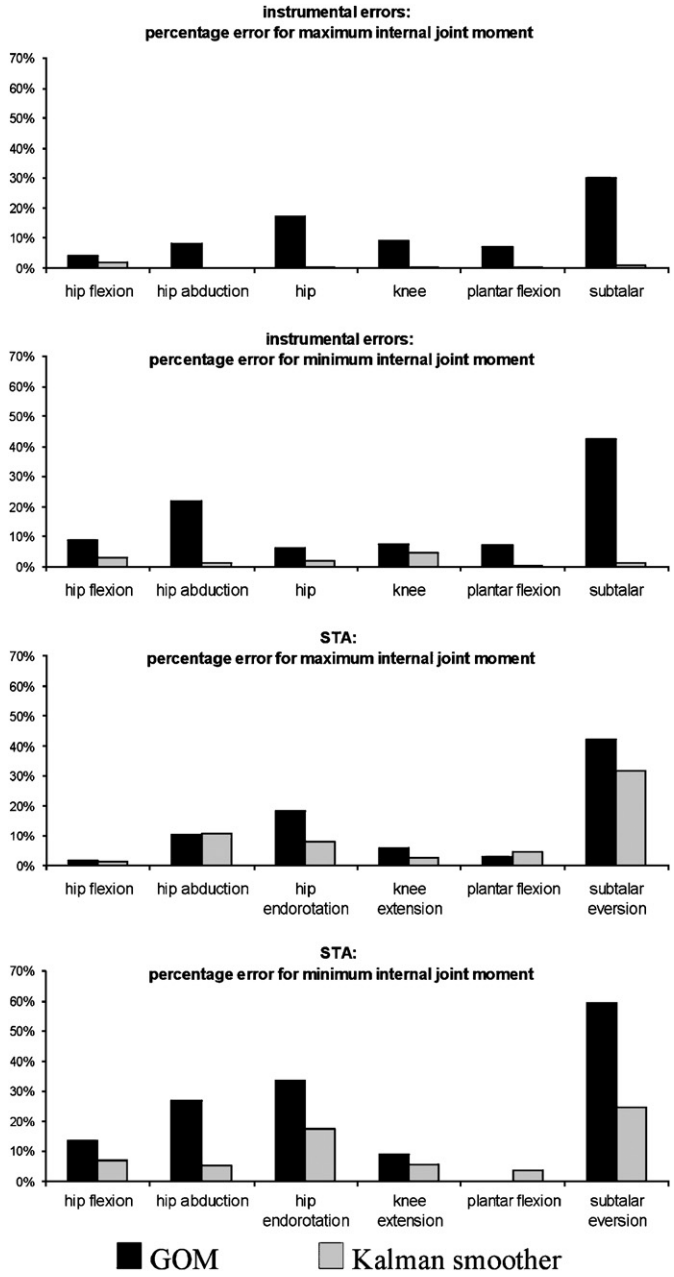


Fig. 5. Percentage errors for the minimum and maximum internal joint moments of the right leg (instrumental errors: above, STA: below) using the GOM and KS.

predicted additional validation trajectories than the estimates obtained with GOM (Table 5). In accordance with the simulation results, the quality of the position estimates was similar for filtering and smoothing. However, the simulation study shows that KF and KS differed for the acceleration estimates. Accelerations were not measured experimentally and consequently no validation at this level was performed.

The Kalman smoother is based on a complex biomechanical model with 21 DOFs. Our implementation however, is not model-specific. Any model can be used in which the joint kinematics is described as a user-defined sequence of three orthogonal translations and three rotations around user-defined axes, which are either constant or a function of the generalized co-ordinates.

Parameters of process and measurement noise were determined based on the expected (K+1)st derivative of the generalized co-ordinates and the expected errors in the measurement of the

Table 5
RMS distance between the marker trajectories calculated based on the estimates of the generalized co-ordinates obtained with the global optimisation method (GOM), Kalman filtering and Kalman smoothing, and the measured marker trajectories of three thigh markers.

$\bar{d}_{RMS}$ (mm)	GOM	Kalman filter ($K=3$)	Kalman smoother ($K=3$)
Estimation marker	12.34	10.34	10.33
Validation marker 1	15.26	11.57	11.55
Validation marker 2	17.17	14.55	14.51

Estimation was based on a marker set, including only one marker on the thigh (estimation marker). The other two markers on the thigh (validation marker) were used for validation.

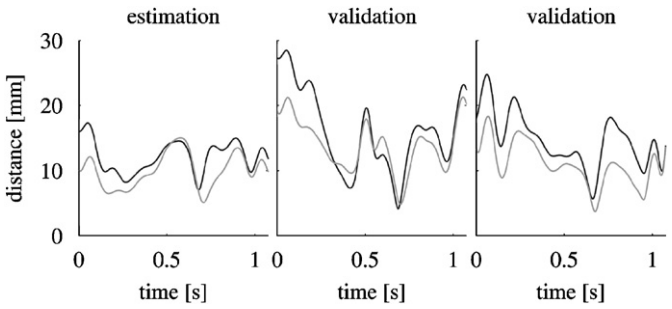


Fig. 6. RMS distance between the marker trajectories calculated based on the estimates of the generalized co-ordinates obtained with the GOM (black) and third-order KS (gray), and the measured marker trajectories for the thigh marker used for estimation (left) and the two validation markers (middle, right).

marker positions, respectively. Varying these parameters over an order of magnitude did not influence the results significantly. In contrast to Cerveri et al. (2005), we found that accurate tuning of these parameters was not crucial.

The main limitation of third-order KS is that STA and modelling errors are not yet adequately handled. KS is a stochastic technique, in which the stochastic model noise accounts for modelling and measuring errors. Both STA and modelling errors are deterministic and consequently not adequately handled. Cerveri et al. (2005) already mentioned that it is impossible to discriminate between a translation of the segment and STA, if all markers on the segment undergo identical STA. We feel that effective handling of STA requires additional information, such as a good model of STA during gait. However, such models are not currently available (Leardini et al., 2005).

In conclusion, third-order KS substantially improves the estimation of joint kinematics and kinetics compared with previously proposed techniques, GOM, KF, and LME, for both simulated, with and without modelling errors, and experimentally measured gait motion.

Conflict of interest statement

All authors hereby declare that there are no conflicts of interest.

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Appendix A. Supporting Information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jbiomech.2008.09.035.

References

Bar-Shalom, Y., Li, X., 1993. Estimation and Tracking, Principles, Techniques and Software. Artech House, Norwood, MA, pp. 382–388.

Cappozzo, A., Catani, F., Leardini, A., Benedetti, M.G., Della Croce, U., 1996. Position and orientation in space of bones during movement, experimental artefacts. Clinical Biomechanics 11 (2), 90–100.

Cerveri, P., Pedotti, A., Ferrigno, G., 2003a. Robust recovery of human motion from video using Kalman filters and virtual humans. Human Movement Science 22, 377–404.

Cerveri, P., Rabufetti, M., Pedotti, A., Ferrigno, G., 2003b. Real-time human motion estimation using biomechanical models and nonlinear state-space filters. Medical & Biological Engineering & Computing 41, 109–123.

Cerveri, P., Pedotti, A., Ferrigno, G., 2005. Kinematical models to reduce the effect of skin artifacts on marker-based human motion estimation. Journal of Biomechanics 38, 2228–2236.

Chiari, L., Della Croche, U., Leardini, A., Cappozzo, A., 2005. Human movement analysis using stereophotogrammetry. Part 2: instrumental errors. Gait & Posture 21, 197–211.

Delp, S.L., Loan, J.P., Hoy, M.G., Zajac, F.E., Topp, E.L., Rosen, J.M., 1990. An interactive graphics-based model of the lower extremity to study orthopaedic surgical procedures. IEEE Transactions on Biomedical Engineering 37 (8), 757–767.

Kalman, R.E., 1960. A new approach to linear filtering and prediction problems. Transactions of the ASME, Journal of Basic Engineering 82, 34–35.

Leardini, A., Chiari, L., Della Croce, U., Cappozzo, A., 2005. Human movement analysis using stereophotogrammetry. Part 3. Soft tissue artifact assessment and compensation. Gait & Posture 21, 212–225.

Lu, T.-W., O'Connor, J.J., 1999. Bone position estimation from skin marker co-ordinates using global optimisation with joint constraints. Journal of Biomechanics 32, 129–134.

Moré, J.J., 1977. The Levenberg–Marquardt algorithm: implementation and theory. In: Watson, G.A. (Ed.), Numerical Analysis. Lecture Notes in Mathematics, vol. 630. Springer, Heidelberg, pp. 105–116.

Rauch, H.E., Tung, F., Striebel, C.T., 1965. Maximum likelihood estimates of linear dynamic systems. AIAA Journal 3 (8), 1445–1450.

Spoor, C.W., Veldpaus, F.E., 1980. Rigid body motion calculated from spatial co-ordinates of markers. Journal of Biomechanics 13, 391–393.

Sutherland, D.H., 2002. The evolution of clinical gait analysis. Part II: kinematics. Gait & Posture 16, 159–179.

Yamaguchi, G.T., Zajac, F.E., 1989. A planar model of the knee joint to characterize the knee extensor mechanism. Journal of Biomechanics 22 (1), 1–10.